

Unknown is not normal: separating language-model extraction from rule-based decision logic for clinical risk scores

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Abstract

Background. Large language models (LLMs) are increasingly used to compute clinical risk scores from free-text notes. Notes are often incomplete, and the usual convention of treating undocumented findings as normal can silently misclassify patients. An end-to-end LLM agent can instead ask the clinician for missing data, but it may ask unnecessary questions or answer before it has enough information.

Objective. To test a division of labour in which an LLM only reads the note, labelling each input present, absent or unknown, and deterministic code decides whether the unknowns could change the decision. We asked whether this reduces the questions put to the clinician without losing accuracy or safety, compared with asking for every missing input, with an end-to-end LLM agent, and with the usual convention of treating missing inputs as normal.

Methods.

- *Cohort and notes.* 1,200 synthetic emergency cases, 200 each for HEART, CURB-65, qSOFA, PERC, Wells (pulmonary embolism) and Cockcroft-Gault. Notes were generated by an LLM from structured ground truth and validated for faithfulness.
- *Extraction.* An LLM labelled each input present, absent or unknown, with an exact evidence span and a confidence. Code computed the lowest and highest score still possible given the unknowns (the score's *bounds*) and asked a simulated clinician only about inputs that could change the decision category (the *bounds policy*).
- *Comparators.* Asking for every missing input; the same pipeline with missing inputs treated as normal; an end-to-end agent (Claude Opus 5.5) with calculator and ask-the-clinician tools; and a variant with value-of-information ordering and confirmation of low-confidence values.
- *Conditions.* Clean and messy notes, and an ideal and a noisy clinician (who does not know, misremembers, or answers vaguely).
- *External check.* 684 real case reports from MedCalc-Bench Verified (584 training, 100 test).
- *Outcomes.* Decision-category accuracy (abstention counted as incorrect) and questions per case. Secondary outcomes were under-triage (a lower-risk category than the truth) and over-triage, premature commitment (answering before the category was determined), and irrelevant questions (whose answer could not change the category).

Results.

- *Ideal conditions.* With a small commercial model (Claude Haiku 4.5) as extractor, the bounds policy matched the accuracy of asking for every missing input (99.4% vs 99.4%) while asking half as many questions (0.92 vs 1.78 per case; $p < 0.001$), none of them irrelevant (vs 48%).
- *Missing treated as normal.* This dropped accuracy to 91.2% and under-triaged 8.5% of patients (95% CI 7.1–10.2); the bounds policy under-triaged 0.1%. This under-triage persisted in every condition (8.2–10.2%), even where its overall accuracy matched the bounds policy (noisy clinician).
- *End-to-end agent.* The Opus agent was as accurate under ideal conditions (99.6%) but asked 9.5% irrelevant questions. With a noisy clinician it was less accurate than the bounds policy (83.5% vs 87.0%; $p < 0.001$) and committed prematurely in 2.7% of cases (bounds: 0%).
- *Local model.* With a 9B model running locally as extractor, accuracy equalled that of perfect extraction (99.8%).

- *Real notes.* Only 52% of the 584 real case reports in the MedCalc-Bench training split contained enough information to determine the category (HEART 13%). Missing-equals-normal gave the correct category in 69%.

Conclusions. Treating undocumented findings as normal systematically under-triages patients, and benchmark conventions hide this. Routing decisions through code that reasons explicitly about unknowns removes premature commitment and irrelevant questions, halves the questions asked, and works with small local models. A frontier agent is competitive on accuracy but not on these safety properties.

Plain-language summary. When a note does not mention a finding, AI tools usually assume it is normal. In our tests this put about one patient in twelve in a lower-risk group than they belonged to. A safer design lets the AI only read the note and leaves the arithmetic to ordinary code, which knows what is still unknown and whether it could change the decision. It then asks the clinician only those questions: about one per patient instead of two, with the same accuracy, and it never answers before the answer is settled.

1. Introduction

Clinical decision rules and scores (HEART for chest pain, CURB-65 for pneumonia, qSOFA for suspected sepsis, PERC and the Wells criteria for pulmonary embolism, and Cockcroft-Gault for renal dosing) are defined over a small set of inputs with explicit thresholds. What matters clinically is usually not the exact score but its *decision category*: for example, a HEART score of 0–3 is low risk, 4–6 moderate and 7 or more high. Several groups now use LLMs to compute them from clinical text (Khandekar et al. 2024; Wang et al. 2025; Zhu et al. 2026). The standard benchmark, MedCalc-Bench, scores the numeric answer and, like most implementations, treats an input that the note does not mention as absent or normal (Khandekar et al. 2024).

In emergency practice that convention is not neutral. “No mention of hemoptysis” is not “no hemoptysis”, and a score computed as if it were can place a patient in a lower-risk category than their true one. The safe behaviour is to recognise when the documented facts do not determine the decision, and then to ask only for the facts that could change it. Recent work shows that LLMs handle this poorly in both directions: they commit to a score when the category is still undetermined, and they abstain when it is already determined (Watanabe et al. 2026). Interactive benchmarks show that models asked to gather information before answering often ask too little or too much (Li et al. 2024; Schmidgall et al. 2024; Johri et al. 2025).

We test a simple division of labour. A language model reads the note into typed, three-valued facts: present with value, explicitly absent, or unknown. Each fact carries an exact evidence span and a confidence. Deterministic code then computes the score’s possible values over the unknown inputs (its *bounds*), decides whether the decision category is already determined, and, if not, identifies the unknown inputs whose values could change the category. Only those are asked. We compare this design with asking for every missing input, with the same pipeline using a binary schema (missing = normal), and with an end-to-end frontier LLM agent that has a calculator and an ask-the-clinician tool.

We pre-specified four hypotheses:

- **H1.** The bounds policy asks far fewer questions than ask-all without losing accuracy.
- **H2.** An end-to-end agent both over-asks (asks questions that cannot change the category) and under-asks (answers before the category is determined).
- **H3.** Three-valued typing reduces silent missing-as-absent errors (an input the note does not mention treated as absent) compared with a binary schema.
- **H4.** Confirming low-confidence, decision-critical extracted values catches most extraction errors at a small question cost.

2. Methods

2.1 Calculators and decision categories

Six calculators were implemented as pure functions from their primary sources (Six et al. 2008; Lim et al. 2003; Seymour et al. 2016; Singer et al. 2016; Kline et al. 2004, 2008; Wells et al. 2000; van Belle et al. 2006; Cockcroft

and Gault 1976). Decision categories were:

- HEART: low 0–3, moderate 4–6, high ≥ 7 .
- CURB-65: 0–1, 2, and ≥ 3 .
- qSOFA: positive if ≥ 2 .
- PERC: negative if all criteria are met.
- Wells: two-tier, PE likely if > 4 .
- Cockcroft-Gault: creatinine clearance < 30 , 30 to < 60 , and ≥ 60 mL/min.

Where the primary source was ambiguous, or differed from common implementations, we implemented a documented reading and listed it for review (Supplementary Table S1). Examples are the HEART troponin bands, the BUN equivalent of CURB-65 urea, and the PERC oxygen-saturation threshold. The author, an emergency physician, reviewed these readings.

Units are normalised by code, never by the model. For example, blood urea nitrogen or urea in mg/dL is converted to urea in mmol/L, and creatinine in μ mol/L to mg/dL.

2.2 Score bounds and decision relevance

Each input is boolean, ordinal, or numeric within a physiological range. Numeric inputs used through thresholds are divided into regions by the calculator’s cut-points, with the inclusivity of each cut-point ($\geq$ vs $>$) declared explicitly. The continuous score, Cockcroft-Gault, is monotone in each input.

Given a partial assignment (known values, intervals for inputs documented as “normal”, and unknowns), the achievable scores are enumerated exactly over the regions, or computed at interval corners for the continuous score. This yields the set of possible categories. The category is *determined* when only one category is possible. An unknown input is *decision-relevant* if changing its value changes the category for some completion of the other unknowns. This definition keeps inputs that matter only jointly.

For example, suppose a HEART patient scores 2 points on the documented inputs and troponin (0–2 points) is unknown. The possible scores are 2–4, spanning low (0–3) and moderate (4–6) risk, so the category is undetermined and troponin is decision-relevant. Had the documented inputs scored 0, the possible scores would be 0–2, all low risk, and no question would be needed.

Property-based tests with Hypothesis checked, for every calculator, that every completion of a partial assignment falls within the computed bounds, and that inputs labelled irrelevant never change the category.

2.3 Synthetic cohort

For each calculator we sampled hidden patient truth from plausible emergency-department priors (Supplementary Table S2). The author reviewed the priors. Each input was then marked documented positive, documented negative or normal, or not documented, with 30% missingness per input; systolic and diastolic blood pressure were documented together.

Cases were resampled until exactly half had a category that the documented facts did not determine, so that both situations were well represented. Within each half, cases keep their natural distribution, so results can be reweighted to the share of undetermined cases that occurs without resampling (Supplementary Table S3).

Cases were tagged with five kinds of trap:

- an embedded negation;
- a value reported in another unit;
- a value from a previous encounter;
- a superseded same-visit reading, chosen where possible to change the category if misread;
- a comorbidity implied only by a medication.

Table 1 describes the cohort.

2.4 Clinical notes

Generation. Notes were written by an LLM (Claude Sonnet 5) from a per-case fact sheet. The fact sheet told the model to state each documented value exactly, to state negatives and normals explicitly, and to omit undocumented inputs entirely, including indirect cues such as rhythm descriptors that imply a heart rate, complete medication lists, or gendered wording. The note was never to name the score.

Validation. Every note passed two layers of validation:

- deterministic rule checks: required numbers present in exact surface form, and no score names;
- an independent LLM judge (Claude Opus 5.5), which classified every input as stated, negated or normal, not mentioned, or implied, compared against ground truth.

Notes that failed were regenerated up to twice (22 of 1,200 needed one retry, and all passed). The prompt was fixed at version v3 after two pilot rounds, and all 1,200 main notes used it.

Messy notes. A “messy” variant rendered 600 of the same cases (100 per calculator) from the same fact sheets in an end-of-shift style, with fragments, heavy abbreviation, typos outside numbers, copied-forward text and no headings. Of these, 41 still failed validation after three attempts and were excluded, leaving 559 cases paired across conditions.

2.5 Extraction

Schema. The extractor returned, for every input:

- a status (present, absent or unknown);
- a value in the input’s domain;
- a unit token from a fixed list;
- the shortest exact supporting quote;
- a confidence.

Code-side checks. Code located the quote in the note and rejected any claim whose quote was not an exact substring; the input then became unknown. Code also normalised units, and rejected “absent” for inputs that cannot be negated (age, weight, sex) and values outside plausible ranges.

Extractors.

- An oracle that returns the documented facts exactly, as a perfect-extraction reference.
- Claude Haiku 4.5, accessed through headless Claude Code (`claude -p`) with a replaced system prompt, all tools disabled and JSON-schema-constrained output. Confidence was self-reported.
- Qwen3.5-9B (4-bit quantised), run locally through Ollama with a JSON schema, thinking disabled, temperature 0, and a fixed seed. Confidence was taken from token log-probabilities as $P(\text{status tokens}) \times P(\text{value tokens})$.

Calibration. Temperature scaling and isotonic regression were fitted on a seeded 30% development split per calculator and evaluated on the remainder. S4 used isotonic calibration.

2.6 Policies

We compared five strategies, labelled S1–S4 and S3-bin in the tables:

- **S1 ask-all:** asks for every input the extraction left unknown.
- **S3 bounds:** asks only decision-relevant unknowns, one at a time, recomputing bounds after each answer.
- **S4:** S3 plus value-of-information ordering (the probability, under population priors, that the answer alone determines the category) and confirmation of extracted values whose calibrated confidence was below 0.9 and that were decision-critical.
- **S3-bin:** S3 with missing inputs treated as absent or normal. This is the MedCalc-Bench convention.
- **S2 end-to-end agent:** Claude Opus 5.5 received the note and parameter list and returned one JSON action per turn: ask the clinician about an input, run the calculator on supplied values, or answer with a category or “cannot determine”.

A code policy abstains when no remaining question can determine the category. For secondary analyses, an abstention is replaced by the highest-risk category still possible. For the agent, which does not track bounds, the calculator’s highest-risk category is used.

2.7 Simulated clinician

A deterministic simulator answered from the hidden truth. Troponin was “not yet available” in 10% of cases. The noisy clinician, drawn independently per case and input, was:

- unable to answer 10% of questions;
- wrong 5% of the time (a yes/no flipped, a graded input one level off, or a number off by 10–25%);
- vague for 25% of numeric answers (a $\pm 10\%$ range containing the truth).

The noise does not depend on question order, so all policies face the same clinician.

2.8 Outcomes and statistics

Primary outcomes. Decision-category accuracy, with abstention counted as incorrect, and questions per case.

Secondary outcomes:

- accuracy when committing;
- under-triage (decision lower-risk than truth) and over-triage (decision higher-risk than truth), using the fallback;
- premature commitment (answering while the category was undetermined by the documented facts plus answers received);
- irrelevant questions (asked when the input could not change the category);
- silent missing-as-absent claims (an input the note does not mention labelled absent);
- extraction accuracy by documented state. A comorbidity conveyed only through a medication (e.g. lisinopril, never naming hypertension) is not determined by the note, so both “present” and “unknown” are accepted there, and “absent” is an error. For the same reason, the reference knowledge used to judge the agent’s questions and premature commitment treats such comorbidities as unknown;
- calibration (Brier score, expected calibration error).

Tests. Proportions have Wilson 95% confidence intervals and mean questions have 2,000-resample bootstrap intervals. Systems were compared on the same cases with exact McNemar tests (accuracy) and Wilcoxon signed-rank tests (questions). P-values were Holm-adjusted within each condition across four pre-specified comparisons (S3 vs S1, S3 vs S2, S3 vs S3-bin, S4 vs S3). Errors were attributed to abstention, extraction, routing (premature commitment), clinician error, or bounds.

2.9 External check on real notes

We used the MedCalc-Bench Verified release ([Khandekar et al. 2024](#)), with its label audits ([Ye et al. 2025](#); [Krohn-Grimberghe 2026](#)), for the five overlapping calculators (qSOFA is not included). This comprised 100 test notes and a seeded sample of 584 training notes, with duplicates of test notes excluded. These are published, de-identified case reports licensed CC-BY-SA 4.0. They were used under a documented exception to the project’s synthetic-only rule and were not redistributed.

For each note we report:

- whether our calculator, applied to MedCalc’s annotated inputs, reproduces their label;
- the share of notes whose category our pipeline found determined;
- whether the true category remained within our bounds;
- category accuracy under MedCalc’s missing-equals-normal convention.

2.10 Implementation and reproducibility

Cost and model access. All runs used \$0 of API spend. Claude models were accessed through a Claude Code subscription, and Qwen ran on a laptop (Apple M1 Pro, 16 GB).

Reproducibility. Every run is driven by one configuration file and a fixed seed. Every model call was cached on disk under a hash of provider, model, prompt and parameters, so the author's tables and figures regenerate from that cache without further model calls (calc-bounds paper configs/main.yaml). The cache is not released; re-running the pipeline re-queries the same models with the same prompts.

Code and tests. The code (Python 3.12, MIT licence; <https://github.com/nicoveraz/calc-bounds>) has 150 automated tests, including the property tests.

3. Results

3.1 Ideal conditions (Table 2, Figure 2)

H1 (supported). With oracle extraction, the bounds policy (S3) matched ask-all (S1) accuracy (99.8% vs 99.8%) with 0.92 vs 1.74 questions per case (Wilcoxon $p < 0.001$). Of S1's questions, 47% could not change the decision; S3 asked none. The pattern held with both LLM extractors:

- Haiku: 99.4% for both, 0.92 vs 1.78 questions.
- Qwen: 99.8% for both, 1.23 vs 2.21 questions.

Across missingness levels of 10%, 30% and 50%, reweighted to the natural share of undetermined cases, S3 needed 34–49% fewer questions than S1 at equal accuracy (Supplementary Table S3).

H3 (supported). Treating missing inputs as normal (S3-bin) reduced accuracy to 91.2–91.8% (McNemar $p < 0.001$ vs S3; 99–105 vs 2 discordant cases). S3-bin committed while the category was undetermined in 39–49% of cases and under-triaged 8.2–8.8% of patients. S3 under-triaged 0.0–0.1%.

H2 (partly supported), ideal conditions. The Opus agent reached 99.6% accuracy with 0.99 questions per case. It was not significantly different from S3 in accuracy, and it asked more questions ($p < 0.001$). Its behaviour included:

- 9.5% of its questions could not change the decision;
- it answered prematurely in only 0.2% of cases, so under-asking was rare here; it became more frequent with a noisy clinician (Section 3.2).

H4 (partly supported). Confirming low-confidence values (S4) raised Haiku-based accuracy from 99.4% to 99.7% and asked fewer questions overall than S3 (0.90 vs 0.92; $p = 0.005$), because value-of-information ordering offset the confirmation questions. The accuracy gain was not significant (3 vs 0 discordant cases, adjusted $p = 0.75$), because Haiku made few extraction errors to catch.

Extractors. Both were accurate and conservative:

- *Accuracy of claims.* 99.6% of Haiku's and 99.5% of Qwen's present/absent claims were correct.
- *How Qwen fails.* Qwen missed 17% of documented negatives, but it labelled them unknown rather than absent, and S3 recovered them by asking.
- *How Haiku fails.* Haiku converted an undocumented input into "absent" in 16 claims (0.022 per case). This is the error H3 targets, and it accounts for most of Haiku's accuracy loss under S3.
- *Calibration.* Isotonic calibration reduced expected calibration error from 0.014 to 0.002 (Haiku) and from 0.048 to 0.002 (Qwen).

3.2 Harder conditions (Table 3, Figures 2–3)

Noisy clinician. A noisy clinician reduced the accuracy of every system that asks questions (S1–S4 and the agent) by 12–16 points, mainly through abstentions: a "don't know" to the one decisive question leaves the category open. When committing, S1, S3 and S4 were 96.8–97.8% correct.

H2, noisy clinician (full cohort, Haiku extraction):

- The agent was less accurate than S3 (83.5% vs 87.0%; 23 vs 65 discordant cases; adjusted $p < 0.001$).
- It committed prematurely in 2.7% of cases; the code policies did so in none.

The binary schema under noise. S3-bin's headline accuracy (86.6–88.4%) matched S3's, because it rarely asked and so rarely met a “don't know”. When it committed, however, it was correct only 89–91% of the time, and it continued to under-triage 8.4–10.2% of patients (Figure 3).

Under- and over-triage with the fallback.

	S3 bounds	S3-bin	Agent
Under-triage, all conditions	0.0–0.5%	8.2–10.2%	up to 1.1%
Over-triage, noisy clinician	9.6–12.9% (S3 and S4)	2.6–3.8%	12.0–13.8%

S3-bin over-triaged least because it rarely left a case undetermined; the price was under-triage. S3 and S4 over-triaged when information was genuinely unavailable, which is the safe direction.

Messy notes (559 paired cases, ideal clinician).

- Haiku's S3 accuracy fell from 99.1% on clean notes to 98.6% on messy notes, and its silent missing-as-absent rate doubled (0.045 per case). S4 recovered part of this (99.1%).
- Qwen was unaffected (99.5%).
- The agent reading messy notes directly was more accurate than S3 with Haiku extraction (99.8% vs 98.6%; 7 vs 0 discordant cases; adjusted $p = 0.047$) and similar to S3 with Qwen extraction.
- With a noisy clinician on messy notes, S3 and the agent did not differ (85.5% vs 85.0%).

3.3 Real notes (Table 4, Figure 4)

Our code reproduced MedCalc-Bench's labels from its annotated inputs for 100% of HEART, CURB-65, PERC and Wells notes. Cockcroft-Gault reproduced only 51% of labels, because MedCalc-Bench chooses the weight input by body-mass index (actual, ideal or adjusted), whereas we used actual weight. One test-split HEART disagreement came from counting a transient ischaemic attack as atherosclerotic disease.

On 584 real training-split case reports:

- Only 52% (95% CI 48–56) had a category determined by the documented facts with Haiku extraction, and 47% with Qwen.
- By calculator: PERC 90%, Cockcroft-Gault 88%, CURB-65 49%, Wells 19%, HEART 13%.
- Under MedCalc's missing-equals-normal convention, the category was correct for 69% of notes (HEART 22%).
- The true category remained possible within our bounds for 93–95% of notes. The misses were extraction errors, mainly in HEART inputs (agreement with annotations 73–74%).

The 100-note test split gave similar results: 59% determined, and 97% with the truth still possible.

4. Discussion

4.1 Principal findings

The most consistent finding is a safety one. Treating undocumented findings as normal under-triaged about one patient in twelve in every condition we tested, including clean notes and a perfect clinician. On real case reports it produced the correct category for only about two-thirds of notes. Headline accuracy can hide this: under a noisy clinician the binary schema's accuracy equalled that of the bounds policy, because a system that rarely asks rarely meets an unanswerable question.

Separating extraction from decision logic removed premature commitment and irrelevant questions entirely, and halved the number of questions compared with asking for everything. When information was genuinely unobtainable, the bounds policy defaulted to the higher-risk category.

A frontier end-to-end agent was a strong baseline. Under ideal conditions it was as accurate as the bounds policy, and on messy notes it was more accurate than the pipeline with a small commercial extractor. We therefore do not claim that the modular design is more accurate than a capable agent. Its advantages lie elsewhere:

- it made no premature commitments and asked no irrelevant questions;
- it had the lowest under-triage;
- it was deterministic and auditable;
- the model only has to read the note, because the decision logic is in code, so a 9B model running locally matched perfect-extraction accuracy. The agent, by contrast, needs a frontier model at every turn (about 2.5 turns per case).

4.2 Relation to prior work

ClinDet-Bench ([Watanabe et al. 2026](#)) formalises the same determinacy question over score ranges and shows that LLMs both commit prematurely and over-abstain. Our work adds the question-asking step and a comparison with an agent that can ask. MediQ, AgentClinic and CRAFT-MD ([Li et al. 2024](#); [Schmidgall et al. 2024](#); [Johri et al. 2025](#)) evaluate information-seeking diagnostic dialogue rather than calculators.

MedCalc-Bench and its audits ([Khandekar et al. 2024](#); [Ye et al. 2025](#); [Krohn-Grimberghe 2026](#)) evaluate one-shot score computation. Our real-note analysis suggests that its missing-equals-normal convention substantially overstates how often a score can safely be computed from the note alone. The design connects to active feature acquisition, selective prediction, and partial evaluation of rules over unknown inputs. Healthcare standards for executable guidelines (CQL, FHIR Clinical Practice Guidelines) would be a natural host for the determinacy check.

4.3 Limitations

- **Synthetic notes.** The main cohort’s notes were generated and validated by LLMs from the same family as the Claude extractor. An exploratory analysis of 46 cases with notes from both Sonnet and Qwen suggested a small same-family advantage (+3.2 points, 95% CI 0.4–6.4). Synthetic notes are also more complete and cleaner than real ones, as the real-note analysis shows, so absolute accuracies are optimistic. Differences in question counts and safety are the more transferable results.
- **Simulated clinician.** It follows a simple parametric model and has not been calibrated to real clinicians.
- **Independent priors.** Patient characteristics were sampled independently (apart from blood pressure), which can produce implausible combinations, such as established coronary disease with no risk factors. It does not change scores, but it lowers the face validity of some cases. We flagged 12 such cases (all HEART: established coronary disease with no risk factors, or age under 40 with three or more risk factors). Excluding them left the results unchanged (Supplementary Table S10).
- **Model access.** Claude models were accessed through Claude Code’s headless mode rather than a bare API. This adds about 1,200 tokens of harness context, and their confidence is self-reported.
- **Real-note evaluation.** It does not include asking questions, because no real clinician was available to answer. Its labels have known errors ([Ye et al. 2025](#); [Krohn-Grimberghe 2026](#)).
- **Calculator readings.** The readings (Supplementary Table S1) and the population priors were reviewed by a single physician, the author, not by an independent panel. Other readings would change some scores, but they apply equally to all strategies.
- **Near-ceiling accuracy.** Accuracy under ideal conditions leaves little room for differences, and several accuracy comparisons are underpowered.

4.4 Implications

For builders of clinical decision support and documentation tools:

- keep “unknown” distinct from “absent” end to end;
- decide with code whether the documented facts determine the decision;
- ask only what can change it;
- when information cannot be obtained, fall back to the higher-risk category.

For evaluators, benchmarks that fill missing inputs with normal values should also report how often the category is actually determined by the note, and how often the convention under-triages.

5. Conclusions

Unknown is not normal. Treating undocumented findings as normal quietly under-triages patients and inflates benchmark accuracy. A simple design in which a language model reads and code decides makes missing information explicit, asks only questions that matter, never commits prematurely, and works with small local models, while matching the accuracy of ask-everything policies and remaining competitive with a frontier agent.

Declarations

- **Data availability.** Only the code is released (see Code availability). The synthetic cohort is regenerated deterministically by the code from the configuration and seed. The synthetic notes, extractions, traces and the LLM response cache are not released; running the code regenerates them with the same prompts, models and seeds, but LLM outputs are not guaranteed to be byte-identical. MedCalc-Bench Verified is available from its authors (CC-BY-SA 4.0).
- **Code availability.** <https://github.com/nicoveraz/calc-bounds> (MIT licence).
- **Funding.** None.
- **Competing interests.** None.
- **Use of AI.** LLMs generated the synthetic notes and served as extractors, judge and agent, as described. Code and analysis were developed with AI assistance (Claude Code).
- **Ethics.** Synthetic data, plus published de-identified case reports; no patient data. Ethics review was not required.

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Figures

Tables

Table 1. Synthetic cohort (n = 1,200). Undetermined: category not determined by the documented facts (50% by design).

Calculator	Cases	Parameters	True categories	Undetermined from note	Params not documented, %	Cases with traps
HEART	200	11	low 82, moderate 103, high 15	100	25	56
CURB-65	200	6	low 95, moderate 68, high 37	100	24	61
qSOFA	200	3	negative 154, positive 46	100	33	53

Calculator	Cases	Parameters	True categories	Undetermined from note	Params not documented, %	Cases with traps
PERC	200	8	negative 86, positive 114	100	29	61
Wells PE	200	7	pe_unlikely 162, pe_likely 38	100	29	56
Cockcroft-Gault	200	4	crcl_lt_30 23, crcl_30_to_lt_60 91, crcl_ge_60 86	100	24	39

Table 2. Main results: clean notes, ideal clinician (n = 1,200 per row). S1 ask-all; S2 end-to-end agent (Opus 5.5; reads the notes, so identical across extraction rows); S3 bounds; S4 bounds + value-of-information ordering + confirmation of low-confidence values; S3-bin bounds with missing = normal. Accuracy counts abstention as incorrect; under-triage uses the higher-risk fallback. 95% CIs: Wilson (proportions), bootstrap (questions).

Extraction	System	Accuracy, % (95% CI)	Questions/case (95% CI)	Irrelevant, %	Premature, %	Under-triage, % (95% CI)
Oracle	S1	99.8 (99.4–100.0)	1.74 (1.67–1.82)	47.3	0.0	0.0 (0.0–0.3)
Oracle	S2	99.6 (99.0–99.8)	0.99 (0.92–1.06)	9.5	0.2	0.1 (0.0–0.5)
Oracle	S3	99.8 (99.4–100.0)	0.92 (0.85–0.99)	0.0	0.0	0.0 (0.0–0.3)
Oracle	S4	99.8 (99.4–100.0)	0.87 (0.81–0.93)	0.0	0.0	0.0 (0.0–0.3)
Oracle	S3-bin	91.8 (90.1–93.2)	0.21 (0.19–0.24)	0.0	39.2	8.2 (6.8–9.9)
Haiku 4.5	S1	99.4 (98.8–99.7)	1.78 (1.71–1.86)	48.1	0.0	0.1 (0.0–0.5)
Haiku 4.5	S2	99.6 (99.0–99.8)	0.99 (0.92–1.06)	9.5	0.2	0.1 (0.0–0.5)
Haiku 4.5	S3	99.4 (98.8–99.7)	0.92 (0.86–0.99)	0.0	0.0	0.1 (0.0–0.5)
Haiku 4.5	S4	99.7 (99.1–99.9)	0.90 (0.83–0.96)	0.0	0.0	0.1 (0.0–0.5)
Haiku 4.5	S3-bin	91.2 (89.4–92.6)	0.21 (0.18–0.23)	0.0	40.0	8.5 (7.1–10.2)
Qwen3.5-9B (local)	S1	99.8 (99.4–100.0)	2.21 (2.12–2.29)	44.3	0.0	0.0 (0.0–0.3)
Qwen3.5-9B (local)	S2	99.6 (99.0–99.8)	0.99 (0.92–1.06)	9.5	0.2	0.1 (0.0–0.5)
Qwen3.5-9B (local)	S3	99.8 (99.4–100.0)	1.23 (1.15–1.31)	0.0	0.0	0.0 (0.0–0.3)
Qwen3.5-9B (local)	S4	99.8 (99.4–100.0)	1.20 (1.13–1.27)	0.0	0.0	0.0 (0.0–0.3)
Qwen3.5-9B (local)	S3-bin	91.2 (89.5–92.7)	0.26 (0.23–0.29)	0.0	49.2	8.8 (7.3–10.5)

Table 3. Harder conditions (notes / clinician), Haiku 4.5 extraction, 559 paired cases; systems as in Table 2 (full grid including Qwen3.5-9B in Supplementary Table S6).

Condition	System	Accuracy, % (95% CI)	Correct if answered, %	Under-triage, % (95% CI)	Over-triage, %	Premature, %	Questions/case
clean / ideal	S1	99.1 (97.9–99.6)	99.3	0.2 (0.0–1.0)	0.7	0.0	1.71 (1.60–1.81)

Condition	System	Accuracy, % (95% CI)	Correct if answered, %	Under-triage, % (95% CI)	Over- triage, %	Premature, %	Questions/case
clean / ideal	S2	99.6 (98.7–99.9)	99.8	0.2 (0.0–1.0)	0.2	0.2	0.92 (0.82–1.01)
clean / ideal	S3	99.1 (97.9–99.6)	99.3	0.2 (0.0–1.0)	0.7	0.0	0.85 (0.75–0.94)
clean / ideal	S4	99.6 (98.7–99.9)	99.8	0.2 (0.0–1.0)	0.2	0.0	0.82 (0.73–0.91)
clean / ideal	S3- bin	91.4 (88.8–93.5)	91.4	8.2 (6.2–10.8)	0.4	37.0	0.20 (0.16–0.24)
clean / noisy	S1	88.2 (85.3–90.6)	97.6	0.5 (0.2–1.6)	8.8	0.0	1.71 (1.60–1.81)
clean / noisy	S2	84.4 (81.2–87.2)	97.7	0.5 (0.2–1.6)	13.1	1.8	0.95 (0.84–1.05)
clean / noisy	S3	88.2 (85.3–90.6)	97.6	0.5 (0.2–1.6)	8.8	0.0	0.88 (0.77–0.97)
clean / noisy	S4	88.7 (85.8–91.1)	98.0	0.5 (0.2–1.6)	8.2	0.0	0.86 (0.76–0.96)
clean / noisy	S3- bin	88.7 (85.8–91.1)	91.2	8.2 (6.2–10.8)	2.5	36.0	0.20 (0.16–0.25)
messy / ideal	S1	98.6 (97.2–99.3)	98.7	0.5 (0.2–1.6)	0.9	0.0	1.92 (1.80–2.04)
messy / ideal	S2	99.8 (99.0–100.0)	100.0	0.0 (0.0–0.7)	0.2	0.7	0.92 (0.82–1.01)
messy / ideal	S3	98.6 (97.2–99.3)	98.7	0.5 (0.2–1.6)	0.9	0.0	1.04 (0.94–1.15)
messy / ideal	S4	99.1 (97.9–99.6)	99.3	0.5 (0.2–1.6)	0.4	0.0	1.04 (0.94–1.15)
messy / ideal	S3- bin	90.7 (88.0–92.8)	90.7	8.8 (6.7–11.4)	0.5	39.4	0.30 (0.23–0.36)
messy / noisy	S1	85.5 (82.3–88.2)	96.8	0.5 (0.2–1.6)	11.1	0.0	1.92 (1.80–2.04)
messy / noisy	S2	85.0 (81.8–87.7)	97.3	1.1 (0.5–2.3)	12.0	3.4	0.97 (0.87–1.08)
messy / noisy	S3	85.5 (82.3–88.2)	96.8	0.5 (0.2–1.6)	11.1	0.0	1.08 (0.97–1.20)
messy / noisy	S4	85.9 (82.7–88.5)	97.4	0.5 (0.2–1.6)	10.7	0.0	1.11 (1.00–1.23)
messy / noisy	S3- bin	86.8 (83.7–89.3)	90.3	8.6 (6.5–11.2)	3.8	38.3	0.31 (0.25–0.38)

Table 4. Real case reports (MedCalc-Bench Verified, training split sample; test split in Supplementary Table S8).

Extractor	Calculator	n	Code reproduces label, %	Determined from note, % (95% CI)	Truth still possible, %	Category correct, missing=normal, %
Haiku 4.5	Cockcroft- Gault	125	51	88.0 (81.1–92.6)	85	71
Haiku 4.5	CURB-65	85	100	49.4 (39.0–59.8)	93	82

Extractor	Calculator	n	Code reproduces label, %	Determined from note, % (95% CI)	Truth still possible, %	Category correct, missing=normal, %
Haiku 4.5	HEART	125	100	12.8 (8.0–19.8)	90	22
Haiku 4.5	PERC	124	100	90.3 (83.8–94.4)	100	94
Haiku 4.5	Wells PE	125	100	19.2 (13.3–27.0)	98	81
Haiku 4.5	All	584	90	52.1 (48.0–56.1)	93	69
Qwen3.5-Cockcroft-9B	Gault	125	51	84.8 (77.5–90.0)	90	72
Qwen3.5-CURB-65 9B		85	100	41.2 (31.3–51.8)	95	81
Qwen3.5-HEART 9B		125	100	12.0 (7.4–18.9)	93	9
Qwen3.5-PERC 9B		124	100	87.9 (81.0–92.5)	100	91
Qwen3.5-Wells PE 9B		125	100	9.6 (5.6–16.0)	99	78
Qwen3.5-All 9B		584	90	47.4 (43.4–51.5)	95	65

Supplementary material: calculator review items (S1), priors (S2), missingness sensitivity with natural-share reweighting (S3), confirmation threshold sweep (S4), calibration (S5), full result grids and paired comparisons (S6–S7), real-note test split (S8), exact prompts (S9).

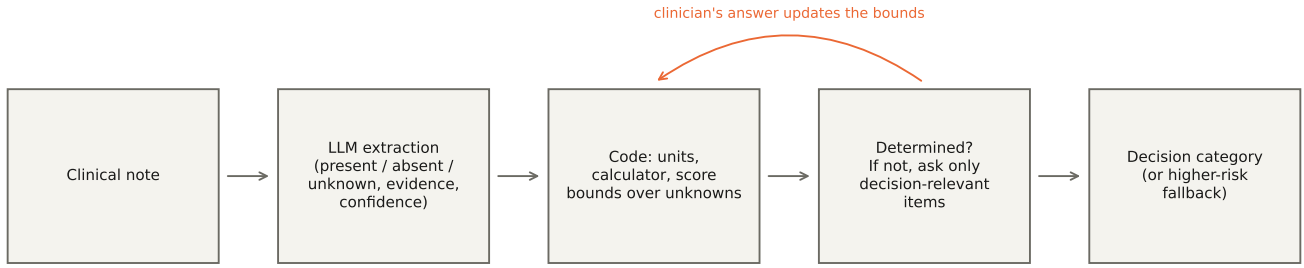


Figure 1: Pipeline. A language model reads the note into typed tri-state facts (present / absent / unknown, with an exact evidence span and a confidence). Code normalizes units, computes the score’s bounds over unknown inputs and, if the decision category is not determined, asks the clinician only for inputs that could change it; each answer updates the bounds.

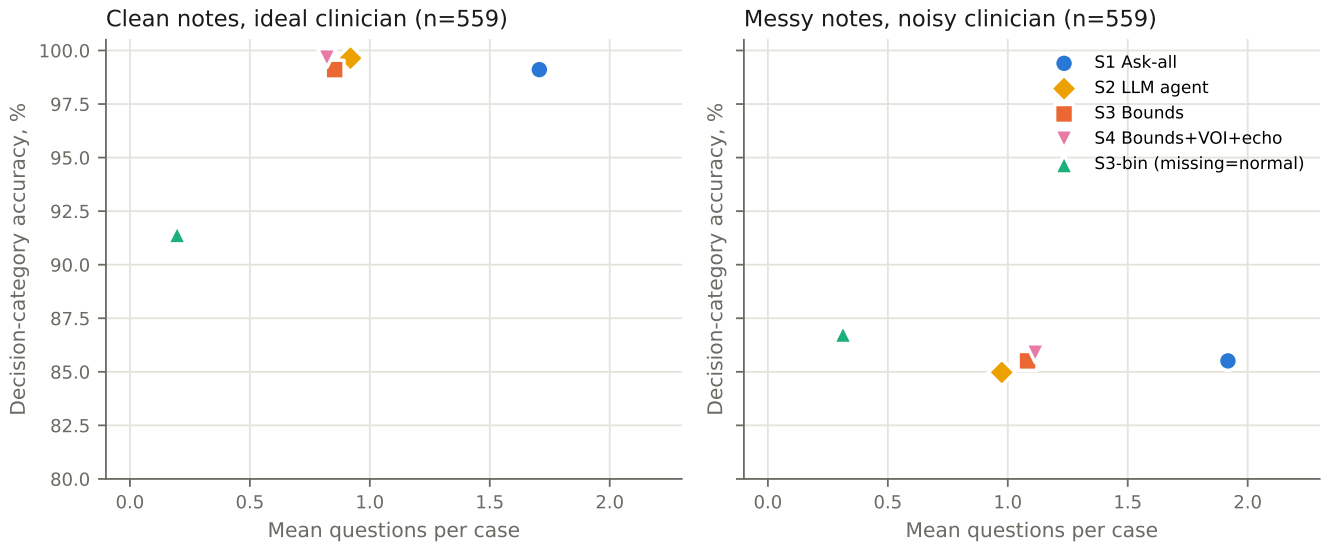


Figure 2: Decision-category accuracy (abstention counted as incorrect) versus mean questions per case, Haiku 4.5 extraction, 559 cases with validated notes in both styles. Left: clean notes, ideal clinician. Right: messy notes, noisy clinician.

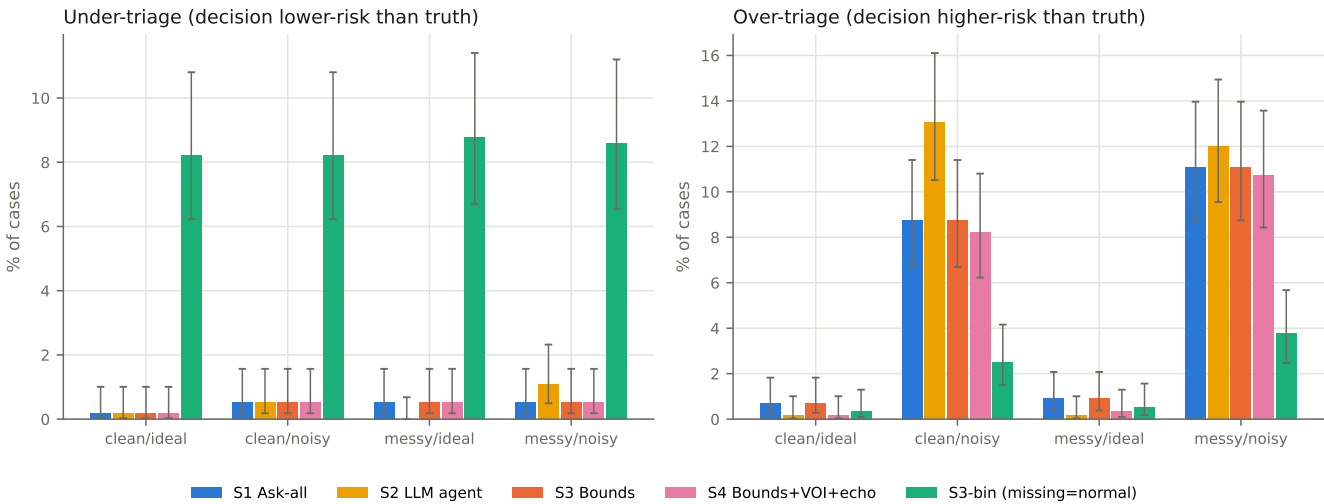


Figure 3: Under-triage (decision lower-risk than the truth) and over-triage (higher-risk) with a higher-risk fallback for undetermined cases, by system and condition (Haiku 4.5 extraction, 559 paired cases). Error bars: Wilson 95% CIs.

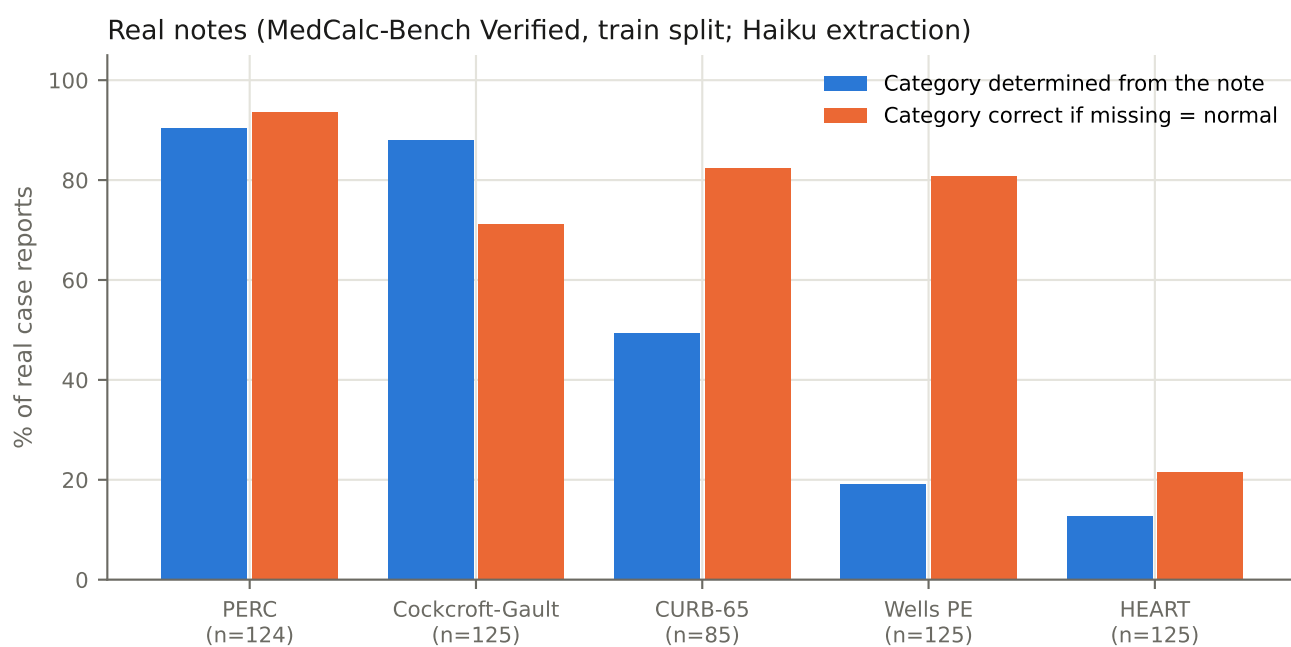


Figure 4: Real case reports (MedCalc-Bench Verified training split; Haiku 4.5 extraction): share whose decision category is determined by the documented facts, versus share whose category is correct when missing inputs are treated as normal.